Worksheet 3.2 — Right-Triangle Trigonometry

Mathematics: Analysis & Approaches SL (IB) · Unit 3: Geometry and Trigonometry

The Big Ideas

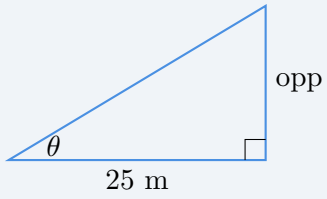
Multi-step right-triangle problems often chain two triangles that share a common side.

- $\sin\theta = \dfrac{\text{opp}}{\text{hyp}}$, $\cos\theta = \dfrac{\text{adj}}{\text{hyp}}$, $\tan\theta = \dfrac{\text{opp}}{\text{adj}}$.
- Two given angles of elevation from different positions usually mean two right triangles sharing the unknown height.

Worked Examples

Example 1 — Angle of elevation with eye height

A person whose eyes are 1.7 m up stands 25 m from a tower and sees its top at an elevation angle of 32ř. Find the tower's height.



Solution. $x = 25\tan(32ř) \approx 15.6$ m; add eye height.

Conclusion: height ≈ 17.3 m.

Example 2 — Two-step angle of depression

From a 60 m cliff, the angle of depression to a boat is 22ř. It sails away until the angle of depression is 11ř. Find the distance sailed.

Solution. $d_1 = \dfrac{60}{\tan 22ř} \approx 148.5$; $d_2 = \dfrac{60}{\tan 11ř} \approx 308.6$.

Conclusion: distance ≈ 160.1 m.

Example 3 — Flagpole on a building

A 5 m flagpole sits atop a building. From the ground, the angle of elevation to the pole's base is 38ř, and to its top is 44ř. Find the building's height.

Solution. $h\tan(44ř) = (h+5)\tan(38ř)$ combined via d; solving $h = \dfrac{5\tan(38ř)}{\tan(44ř) - \tan(38ř)}$.

Conclusion: $h \approx 18.5$ m.

Example 4 — Bearing components

A hiker walks 10 km on a bearing of 055ř. Find how far north and east she travels.

Solution. North: $10\cos 55ř \approx 5.74$; East: $10\sin 55ř \approx 8.19$.

Conclusion: ≈ 5.74 km north, ≈ 8.19 km east.

Example 5 — Perpendicular-bearing distance

Two hikers start together. Hiker A walks 8 km on bearing 070ř; Hiker B walks 6 km on bearing 160ř. Find the distance between them (note the bearings differ by exactly 90ř).

Solution. Since the paths are perpendicular, use Pythagoras directly: $d = \sqrt{8^2 + 6^2} = \sqrt{100}$.

Conclusion: $d = 10$ km.

Example 6 — Two objects from an elevated viewpoint

From a helicopter at height 500 m, the angles of depression to two ships in a line are 25ř and 40ř. Find the distance between the ships.

Solution. $d_1 = \dfrac{500}{\tan 25ř} \approx 1072.3$; $d_2 = \dfrac{500}{\tan 40ř} \approx 595.9$.

Conclusion: distance ≈ 476.4 m.

Example 7 — Height from a shadow

A pole casts a 12 m shadow when the sun's elevation angle is 50ř. Find the pole's height.

Solution. $h = 12\tan(50ř)$.

Conclusion: $h \approx 14.3$ m.

Example 8 — Angle of inclination from rise and run

A ramp rises 0.9 m over a horizontal run of 11 m. Find its angle of inclination.

Solution. $\theta = \tan^{-1}\left(\dfrac{0.9}{11}\right)$.

Conclusion: $\theta \approx 4.7ř$.

Example 9 — Resolving a velocity vector

A plane's velocity is 250 km/h on a bearing of 070ř. Find its northward and eastward speed components.

Solution. North: $250\cos 70ř \approx 85.5$; East: $250\sin 70ř \approx 234.9$.

Conclusion: ≈ 85.5 km/h north, ≈ 234.9 km/h east.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Find the angle of elevation to the top of a 30 m tower from a point 45 m away.

Question 2

A person with eyes at 1.6 m stands 20 m from a tree, seeing its top at 28ř elevation. Find the tree's height.

Question 3

From an 80 m cliff, the angle of depression to a boat is 15ř. Find the horizontal distance to the boat.

Question 4

A plane flies 250 km on a bearing of 160ř. Find its distance south and east of its start.

Question 5 — Challenge

From two points 50 m apart along a line from a tower's base, elevation angles are 20ř (far) and 35ř (near). Find the tower's height.

Question 6 — Challenge

A 6 m flagpole sits on a building. Ground elevation angles are 35ř to its base and 41ř to its top. Find the building's height.

Question 7

A ladder 8 m long leans against a wall, making a 70ř angle with the ground. Find how high up the wall it reaches.

Question 8

A ship sails 40 km on a bearing of 200ř. Find its distance south and west of its start.

Question 9

A pole casts a 20 m shadow when the sun's elevation is 42ř. Find the pole's height.

Question 10

From a helicopter at height 400 m, angles of depression to two cars in a line are 30ř and 50ř. Find the distance between the cars.

Question 11 — Challenge

Two hikers start together; A walks 9 km on bearing 050ř, B walks 12 km on bearing 140ř. Find the distance between them.

Question 12 — Challenge

A plane's velocity is 300 km/h on bearing 035ř; find its north and east speed components.

Question 13 — Challenge

A ramp rises 1.2 m over a horizontal run of 14 m; find its angle of inclination, then the ramp's actual length.

Question Recap & Answer Key

Question Recap

1. Find the angle of elevation to the top of a 30 m tower from a point 45 m away.
2. A person with eyes at 1.6 m stands 20 m from a tree, seeing its top at 28ř elevation. Find the tree's height.
3. From an 80 m cliff, the angle of depression to a boat is 15ř. Find the horizontal distance to the boat.
4. A plane flies 250 km on a bearing of 160ř. Find its distance south and east of its start.
5. From two points 50 m apart along a line from a tower's base, elevation angles are 20ř (far) and 35ř (near). Find the tower's height.
6. A 6 m flagpole sits on a building. Ground elevation angles are 35ř to its base and 41ř to its top. Find the building's height.
7. A ladder 8 m long leans against a wall, making a 70ř angle with the ground. Find how high up the wall it reaches.
8. A ship sails 40 km on a bearing of 200ř. Find its distance south and west of its start.
9. A pole casts a 20 m shadow when the sun's elevation is 42ř. Find the pole's height.
10. From a helicopter at height 400 m, angles of depression to two cars in a line are 30ř and 50ř. Find the distance between the cars.
11. Two hikers start together; A walks 9 km on bearing 050ř, B walks 12 km on bearing 140ř. Find the distance between them.
12. A plane's velocity is 300 km/h on bearing 035ř; find its north and east speed components.
13. A ramp rises 1.2 m over a horizontal run of 14 m; find its angle of inclination, then the ramp's actual length.

✓ Answer Key

1. $\tan^{-1}(30/45) \approx 33.7$ř
2. $20\tan 28$ř $+ 1.6 \approx 12.24$ m
3. $80/\tan 15$ř ≈ 298.6 m
4. south $= 250\cos 20$ř ≈ 234.9 km, east $= 250\sin 20$ř ≈ 85.5 km
5. $h(1/\tan 20$ř $- 1/\tan 35$ř$) = 50 \Rightarrow h(2.747 - 1.428) = 50 \Rightarrow h \approx 37.9$ m
6. $h = \dfrac{6\tan 35\text{ř}}{\tan 41\text{ř} - \tan 35\text{ř}} \approx \dfrac{4.202}{0.169} \approx 24.9$ m
7. $8\sin 70$ř ≈ 7.52 m
8. angle from south is 20ř; south $= 40\cos 20$ř ≈ 37.6 km, west $= 40\sin 20$ř ≈ 13.7 km
9. $20\tan 42$ř ≈ 18.0 m
10. $d_1 = 400/\tan 30$ř ≈ 692.8; $d_2 = 400/\tan 50$ř ≈ 335.6; distance ≈ 357.2 m
11. bearings differ by 90ř: $d = \sqrt{81 + 144} = \sqrt{225} = 15$ km
12. north $= 300\cos 35$ř ≈ 245.75, east $= 300\sin 35$ř ≈ 172.1 km/h
13. $\theta = \tan^{-1}(1.2/14) \approx 4.9$ř; length $= \sqrt{1.44 + 196} \approx 14.05$ m